

TECHNICAL REPORT

Predictive Analytics Time Series Assignment

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Abstract

This assignment studies and analyzes multiple real-world time series with the aim of investigating their characteristics and reliably forecasting future values. Specifically, the following are examined: (a) annual global land temperature anomalies (globtemp), (b) weekly mortality (cmort) in the United States, (c) monthly rainfall in Zevgolatio, Corinthia (2016–2025), (d) quarterly beer sales of an industrial company (2016–2019), and (e) the monthly unemployment rate in Spain (2022–2025). For each time series, diagnostic checks for stationarity, seasonality, and trend are applied, while appropriate models (AR, ARIMA, SARIMA) are fitted, evaluated on the basis of AIC/BIC criteria, and used to produce forecasts with confidence intervals. The results highlighted the importance of proper preliminary analysis for model selection and forecast reliability, while also revealing limitations of classical statistical models when dealing with highly volatile data or extreme values.

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1 Introduction

Time series analysis is one of the most important tools in statistics and data science, with broad applications in economics, meteorology, health, and technology [5]. In this assignment, real-world time series datasets are examined with the aim of understanding their dynamics, detecting trends and seasonality, and forecasting future values through suitable stochastic models such as ARIMA and SARIMA. Preliminary analysis techniques, stationarity checks, and diagnostic tests for the selected models are presented, with particular emphasis on evaluating their forecasting ability.

The structure of the assignment begins with a brief theoretical overview of the main concepts, followed by a detailed description of the data and methodology. The appropriate statistical analysis is then applied separately to each time series, with a detailed presentation of the results and conclusions.

2 Literature Review

Time series analysis is a fundamental tool for understanding and forecasting phenomena that evolve over time [10]. A time series is a sequence of measurements recorded at successive points in time and often exhibits dependence between consecutive values [5].

Stationarity: An important prerequisite in time series analysis is stationarity, meaning that statistical characteristics such as the mean and variance remain stable over time [10]. Lack of stationarity, for example due to trend or seasonality, is often addressed through suitable transformations such as differencing.

Seasonality: Many time series exhibit seasonality, that is, periodic patterns that repeat at a specific interval (e.g., annually or monthly) [3]. Seasonality analysis helps identify and remove these periodic effects in order to improve forecasting.

ARIMA/SARIMA Models: ARIMA (AutoRegressive Integrated Moving Average) models are widely used for time series forecasting because they can handle both stationary and non-stationary series [1]. SARIMA (Seasonal ARIMA) models extend ARIMA by incorporating seasonal terms so that the seasonality of the series can also be modeled [11].

The selection and correct application of these models require appropriate diagnostic tests and the use of statistical criteria such as AIC and BIC in order to achieve the most suitable fit to the data [5].

3 Data and Methodology

3.1 Data

This assignment studies and analyzes multiple real-world time series with the aim of investigating their characteristics and reliably forecasting future values. Specifically, the following datasets are examined:

- **Annual global land temperature anomalies (globtemp):** Annual data from 1850 to 2023 recording changes in average global land temperature.
- **Weekly mortality (cmort) in the United States:** Weekly mortality observations that are often used to analyze epidemiological phenomena and forecasting methods.

- **Monthly rainfall in Zevgolio, Corinthia:** Monthly rainfall values for the years 2016–2025, collected from meteorological data for the region.
- **Quarterly beer sales of an industrial company:** Sales data, in millions of bottles, by season (Winter, Spring, Summer, Autumn) for the years 2016–2019.
- **Monthly unemployment rate in Spain:** Monthly unemployment data from Eurostat and INE for the period January 2022 to April 2025.

3.2 Methodology

The time series analysis includes the following steps:

- **Preliminary analysis:** Visualization and statistical examination of stationary or non-stationary behavior using plots, ACF/PACF, and stationarity tests.
- **Decomposition and preprocessing:** Removal of trend and seasonality where necessary through differencing or decomposition (multiplicative/additive decomposition).
- **Application of suitable models:** Fitting AR, MA, ARMA, ARIMA, or SARIMA models according to the characteristics of each series.
- **Diagnostic checking:** Testing residuals for white-noise behavior (Ljung-Box test), evaluating error distributions (Q-Q plots and histograms), and calculating evaluation metrics (MSE, RMSE, MAE, AIC, BIC).
- **Forecasting:** Estimating future values with confidence intervals and visualizing the results.
- **Comparison and conclusions:** Summarizing results, comparing models, and interpreting the findings for each time series.

4 Exercise 1: Time Series Analysis of Unemployment in Spain

4.1 Data Description

For the analysis of the unemployment-rate time series, monthly unemployment data for Spain covering the period from January 2022 to April 2025 were used. The data were collected from official statistical sources such as Eurostat [4] and INE – Instituto Nacional de Estadística [6], providing a total of 40 consecutive monthly observations. Each observation corresponds to the unemployment rate (%) of Spain’s labor force for the respective month.

This period was selected in order to include possible short-term fluctuations, longer-term trends, and the effects of seasonal factors such as the tourism season and winter slowdown. The data form a one-dimensional numerical sequence (time series), where each element corresponds to one month and has the structure: [Year, Month, Unemployment Rate (%)].

Table 1 presents examples of the first and last values of the series, while the following plot (Figure 1) shows the evolution of Spain’s unemployment rate over the period under study.

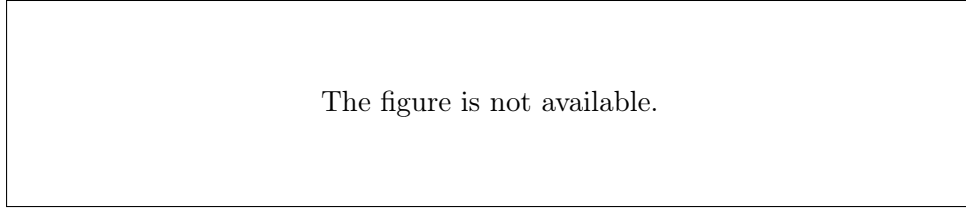


Figure 1: Monthly evolution of the unemployment rate in Spain (Jan. 2022 – Apr. 2025)

Sample data format:

Table 1: Sample Monthly Unemployment Rate Values in Spain (2022–2025)

Year	Month	Unemployment Rate (%)
2022	01	13.6
2022	02	13.8
2022	03	13.8
2022	04	13.1
2022	05	12.5
2022	06	12.4
⋮	⋮	⋮
2025	01	11.2
2025	02	11.5
2025	03	11.5
2025	04	10.9

This time series exhibits pronounced fluctuations, annual seasonality, and a clear downward trend, as discussed in the following sections.

4.2 Time Plot & Seasonality

The evolution of Spain’s unemployment rate from January 2022 to April 2025 shows a strong downward trend, with unemployment decreasing from approximately 13.8% at the beginning of 2022 to 10.9% in April 2025, as shown in Figure 2.

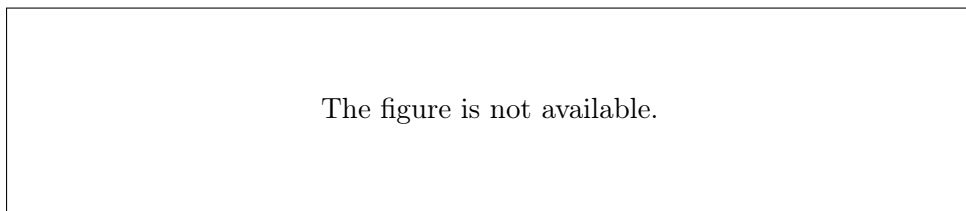


Figure 2: Monthly evolution of the unemployment rate in Spain (Jan. 2022 – Apr. 2025)

This time series is characterized as non-stationary because both its mean and variance change substantially over time, while strong seasonality is also observed. The seasonality

analysis and corresponding plots (Figure 3) reveal a recurring annual pattern, with clear unemployment peaks from January to March and lower levels during May–June.

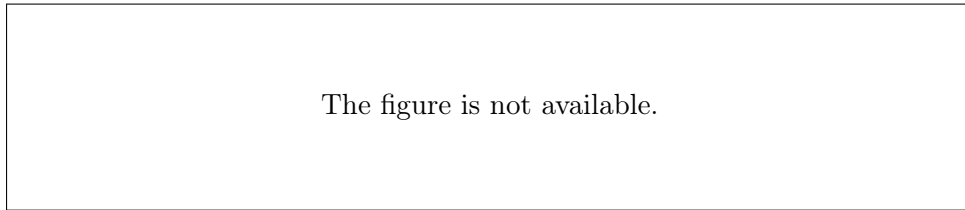


Figure 3: Time series decomposition: trend, seasonality, and random component

This seasonality is confirmed both by the monthly boxplots (Figure 4), which show notable differences between monthly values, and by the autocorrelation function (ACF, Figure 5), which exhibits peaks at 12-month intervals.

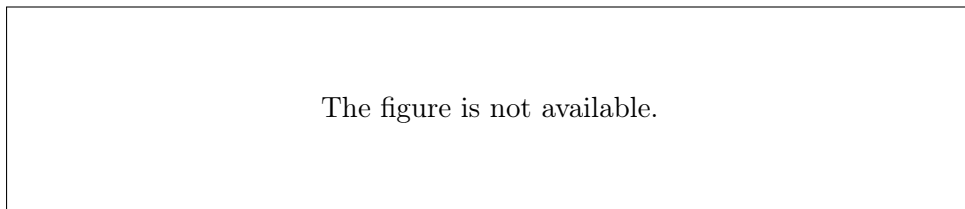


Figure 4: Boxplot of the seasonal variation in the unemployment rate by month

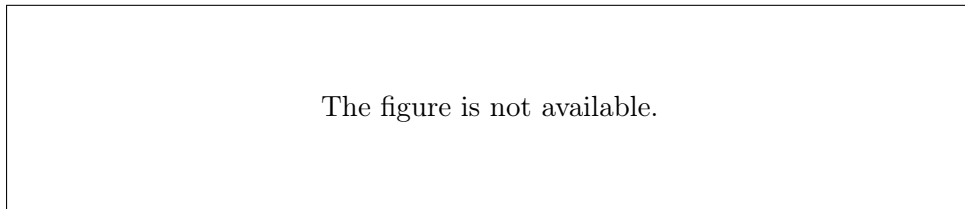


Figure 5: Autocorrelation function (ACF) of the unemployment time series

The observed fluctuations are mainly associated with the seasonal nature of the Spanish economy, as the tourism season increases employment during the summer months, whereas holiday periods and winter are associated with temporary layoffs and higher unemployment.

In conclusion, the analysis of Spain’s unemployment time series reveals both a substantial downward trend and strong annual seasonality, indicating the need for employment policies that take these structural and cyclical factors into account.

4.3 Stationarity Testing & ARMA Fitting

To investigate the stationarity of Spain’s unemployment-rate time series, the autocorrelation (ACF) and partial autocorrelation (PACF) plots were examined. The original ACF and PACF plots showed significant values at many consecutive lags, indicating that the time series is not stationary. First differences were therefore applied to obtain a stationary series. The plot of the first differences (Figure 8) shows that the observations are distributed around zero, a characteristic of a stationary series.

The stationarity of the transformed series is further supported by the ACF of the first differences (Figure 6), where the values decrease rapidly after lag 1, and by the PACF (Figure 7), where one significant value is observed at the first lag. This pattern suggests that the first-differenced series can be appropriately modeled with an ARMA model. ARMA(2,1) and ARMA(1,1) models were tested, using the Akaike Information Criterion (AIC) and residual analysis as selection criteria. The ARMA(2,1) model produced a lower AIC, all coefficients were statistically significant, and its residuals showed white-noise characteristics, confirming a satisfactory fit (Figure 9).

The forecast of the first differences using the selected model for the following months is shown in Figure 10. The predicted values remain close to zero, which is consistent with the stationary nature of the differenced series. Overall, the stationarity procedure and the ARMA(2,1) fit provide a reliable basis for short-term forecasting of the unemployment time series.

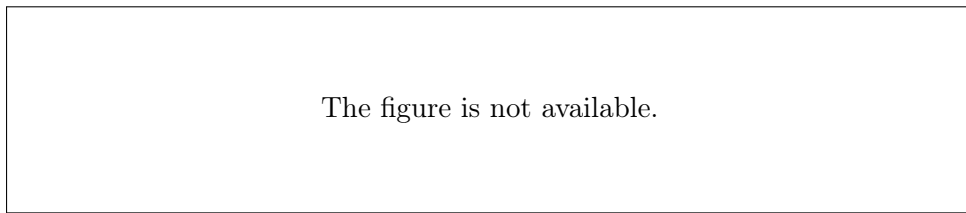


Figure 6: ACF of the first differences of the unemployment time series

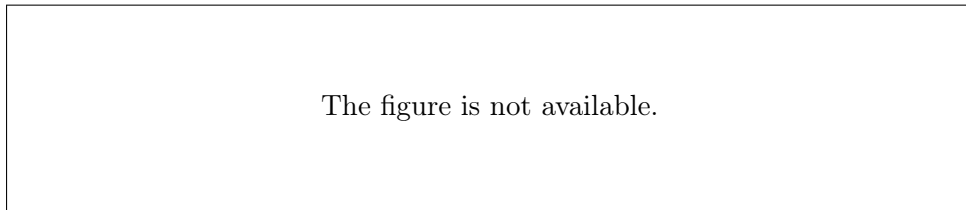


Figure 7: PACF of the first differences of the unemployment time series

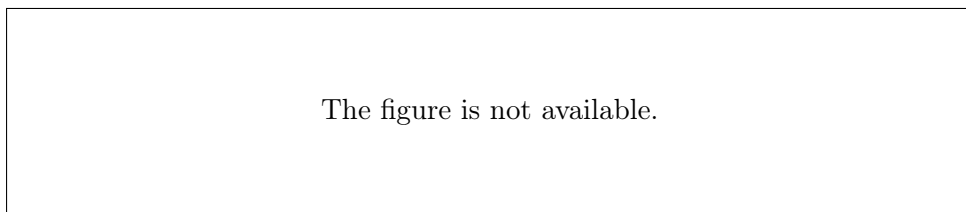


Figure 8: Plot of the first differences of the unemployment rate (stationary series)

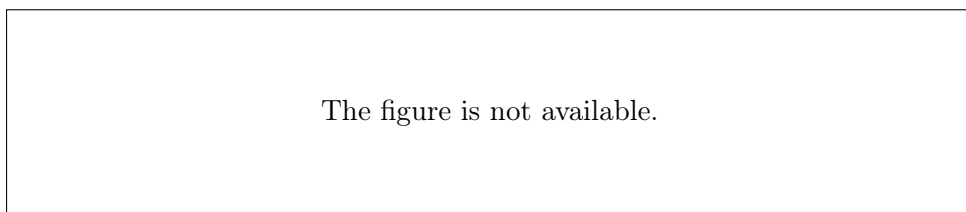


Figure 9: ACF and PACF of the residuals of the fitted ARMA model

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Figure 10: Forecast of the first differences of the unemployment rate using the ARMA(2,1) model

4.4 First Differences and Stationarity Testing of the New Series

Following the finding that the original unemployment-rate time series is non-stationary, the first-difference series was calculated, that is, the difference between each value and the previous one. This series is shown in Figure 8. The autocorrelation (ACF) and partial autocorrelation (PACF) functions were then re-examined for the transformed series (Figures 6, 7). The results show that both the ACF and PACF decay rapidly, indicating that the first-difference series is now stationary. It is therefore suitable for ARMA or ARIMA modeling.

4.5 Fitting and Evaluation of an ARIMA(2,1,1) Model

An ARIMA(2,1,1) model was fitted to the monthly unemployment data for Spain over the analysis period. The estimated model parameters are summarized in Table 2.

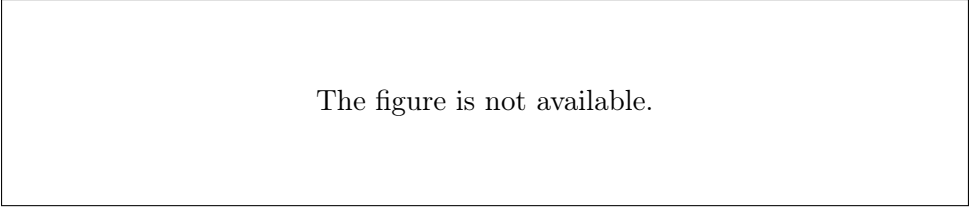
Table 2: Parameter Estimates of the ARIMA(2,1,1) Model

Parameter	Estimate	Std. Error	z-value	p-value
ar1	1.02	0.18	5.64	< 0.001
ar2	-0.68	0.13	-5.29	< 0.001
ma1	-0.32	0.22	-1.46	0.143

Discussion: The ar1 and ar2 coefficients are statistically significant (p -value < 0.05), whereas the ma1 coefficient is not statistically significant, but it is retained in the model to improve the fit.

Residual Diagnostics: Diagnostic examination of the residuals through autocorrelation (ACF) and partial autocorrelation (PACF) plots did not reveal strong autocorrelation, while the Box-Ljung test (p -value = 0.024) indicates borderline statistical significance. This suggests that some weak residual correlation may remain and is not fully captured by the model.

Forecast and Plot: Figure 11 presents the historical unemployment-rate time series (black line), the model forecasts for the following 12 months (red line), and the 95% confidence intervals (blue dashed lines).



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Figure 11: Forecast of unemployment in Spain using the ARIMA(2,1,1) model.

Discussion: The forecast indicates relative stabilization of the unemployment rate over the following year, without extreme fluctuations. The confidence intervals widen as the forecast horizon increases, as expected because uncertainty grows over longer forecasting horizons.

Conclusion: The ARIMA(2,1,1) model provides a satisfactory fit to the data and reliable short-term forecasts for unemployment in Spain. There are minor indications of residual autocorrelation, which could potentially be addressed by testing more complex models or incorporating additional exogenous variables if available.

4.6 Testing for Second Differences

According to the results of Sections 4.4 and 4.5, the first-difference series of the original time series was found to be already stationary, as indicated by the ACF and PACF analyses. Therefore, second differences are not required, nor is further stationarity testing of the transformed series necessary. The stationary first-difference series is considered suitable for ARMA or ARIMA modeling, as applied in the previous sections.

4.7 Final ARIMA Model Fitting to the Original Series

As established from the analysis of the first differences of the unemployment-rate time series, the transformed series is clearly stationary and is therefore appropriate for ARMA or ARIMA models with $d = 1$. Within this framework, an **ARIMA(2,1,1)** model was fitted to the original data, incorporating one difference ($d = 1$). The model results indicate a satisfactory fit and adequate attenuation of residual autocorrelation, while the resulting forecasts exhibit realistic behavior and the expected uncertainty over the forecast horizon. Consequently, the ARIMA(2,1,1) model is considered the most appropriate choice for Spain's unemployment time series over the period examined.

4.8 Model Adequacy Check and Comparison with Extended Models

To evaluate the adequacy of the final **ARIMA(2,1,1)** model, residual analysis was performed and the model was compared with higher-order alternatives.

(I) Residual Check: The autocorrelation (ACF) and partial autocorrelation (PACF) plots of the model residuals (Figure 12) show that the values remain within the significance limits, with no evidence of strong autocorrelation. However, the Box-Ljung test produces a p-value slightly below 0.05 ($p = 0.024$), indicating the possible presence of mild residual structure that is not fully captured by the model.

(II) Comparison with Extended Models: Additional higher-order ARIMA models (ARIMA(3,1,1), ARIMA(2,1,2), ARIMA(2,2,1)) were fitted and compared using the AIC criterion and the Box-Ljung p-value for the residuals. The results are summarized in Table 3.

Table 3: Comparison of ARIMA models in terms of AIC and Box-Ljung p-value.

Model	AIC	Box-Ljung p-value
ARIMA(2,1,1)	13.48	0.024
ARIMA(3,1,1)	12.05	0.071
ARIMA(2,1,2)	5.88	0.0003
ARIMA(2,2,1)	17.79	0.0073

The **ARIMA(2,1,2)** model has the lowest AIC, but the Box-Ljung p-value is extremely small (0.0003), indicating significant autocorrelation in the residuals and therefore an inadequate model. In contrast, the **ARIMA(3,1,1)** model has a relatively low AIC and an acceptable p-value (0.071), indicating better statistical adequacy than the ARIMA(2,1,1) model.

Conclusion: Although the initial ARIMA(2,1,1) model captures the main dynamics of the time series, the extended ARIMA(3,1,1) model provides slightly better residual adequacy. The final model should therefore be selected on the basis of a combination of adequacy measures (AIC, Box-Ljung test) and interpretability, with preference given to ARIMA(3,1,1) when statistical adequacy is the main priority. A low AIC indicates good fit, but the absence of autocorrelation in the residuals is essential for the validity of the model forecasts. Thus, when there is a conflict between AIC and statistical adequacy, the model with better residual behavior is preferred. In statistical time series modeling, final model selection is based not only on fit indices such as AIC, but also on the absence of structure in the residuals so that the assumptions of the model are satisfied.

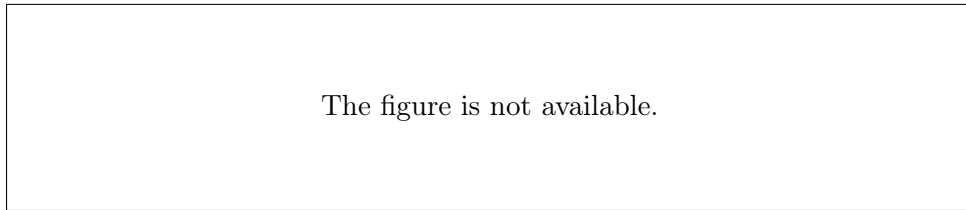


Figure 12: ACF and PACF of the residuals for the ARIMA(2,1,2) model.

4.9 Long-Term Unemployment Forecast

To forecast the future path of Spain's unemployment rate, the final **ARIMA(3,1,1)** model was applied. This model was selected as the most appropriate among several candidate models on the basis of statistical criteria (low AIC and satisfactory residual adequacy).

Figure Figure 13 presents the unemployment-rate forecast for the next 24 months:

- **The black line** represents the historical observed data.
- **The red line** represents the predicted unemployment-rate values based on the selected ARIMA(3,1,1) model.
- **The blue dashed lines** define the 95% confidence interval of the forecasts.

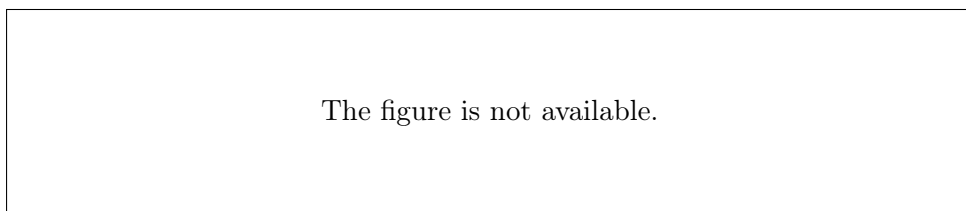


Figure 13: Forecast of the unemployment rate for the next 24 months using the ARIMA(3,1,1) model.

Discussion of Results:

- **Relative stabilization:** The unemployment-rate forecast remains at levels similar to the most recent observations, without showing a strong upward or downward trend.

- **Mild fluctuation:** Small cyclical fluctuations can be observed in the predicted values, possibly reflecting seasonal patterns inherited from the historical sample.
- **Uncertainty:** The 95% confidence intervals widen as the forecast horizon increases, reflecting the normal increase in uncertainty over longer forecasting horizons.
- **Absence of extreme values:** No extreme forecast values or exponential upward/downward trends are observed, suggesting that the model produces conservative and realistic forecasts.

Conclusion: The final **ARIMA(3,1,1)** model provides reliable short- and medium-term forecasts for the unemployment rate, showing relative stability and satisfactory interpretability. Forecast accuracy decreases, as expected, as the forecast horizon increases, which is reflected by the widening confidence intervals.

5 Exercise 2: Analysis of Climate Data

5.1 Data Description

For the analysis of monthly rainfall in Zevgolatio, Corinthia, during the period 2016–2025, open meteorological data obtained from the open-meteo.com platform [9] were used. The original data consisted of daily rainfall values (in millimeters, mm) for each calendar day of the study period.

Initially, the data were imported into the R environment and cleaned so that metadata rows were removed during file reading. The date column (time) was converted to the Date type, and the year and month were then extracted from each record. To create the monthly time series, the daily values were grouped by year and month and the total rainfall for each month was calculated using the aggregate function in R.

The final result is a one-dimensional time series representing monthly rainfall depth (in mm) for each month from January 2016 to June 2025. The following plot (Figure 14) illustrates the evolution of rainfall over the study period.

These data form the basis for the time series analysis and rainfall forecasting for the region.

5.2 Time Plot, Stationarity and Seasonality

Figure 14 presents the time plot of monthly rainfall in Zevgolatio, Corinthia, for the period 2016–2025.

From the plot, the following observations can be made:

- There are large variations in rainfall from year to year, as well as some periods with particularly heavy rainfall. This suggests that the mean and variance of the series change over time; therefore, the time series cannot be considered stationary.
- Repeating within-year patterns are also visible, with higher rainfall during the winter months and lower rainfall during summer. This provides clear evidence of annual seasonality in the series.

In conclusion, monthly rainfall exhibits strong seasonality and non-stationarity, because the statistical properties of the series (mean and variance) vary over time and are influenced by the seasonal climatic conditions of the region.

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Figure 14: Monthly rainfall (in mm) in Zevgolatio, Corinthia, for the period 2016–2025, as calculated from the daily data obtained from open-meteo.com.

5.3 Stationarity Examination Using ACF and PACF

To investigate the stationarity of the monthly rainfall time series for Zevgolatio, Corinthia, the autocorrelation (ACF) and partial autocorrelation (PACF) plots were examined, as shown in Figures 15 and 16.

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Figure 15: Autocorrelation function (ACF) plot of monthly rainfall in Zevgolatio, Corinthia.

The figure is not available.

Figure 16: Partial autocorrelation function (PACF) plot of monthly rainfall in Zevgolatio, Corinthia.

Discussion: The ACF shows a very high value at the first lag, after which the values decrease relatively slowly, with most subsequent lags remaining within the confidence intervals. Similarly, the PACF shows a significant value only at the first lag, while the remaining lags are low.

This pattern, together with the time plot of the series, indicates that the time series is not stationary. The significant autocorrelation at lag 1 and the slow decay of the values are indications of non-stationarity, while the time plot in the previous section also revealed strong seasonality and changes in the mean.

Conclusion: The monthly rainfall time series for Zevgolatio, Corinthia, is not stationary. To achieve stationarity, an appropriate transformation such as first-order differencing and/or seasonal differencing should be applied. Therefore, in its current form the series is not suitable for the direct application of an $\text{ARMA}(p, q)$ model. It must first be transformed into a stationary series, after which the autocorrelation plots should be re-examined to select an appropriate model.

5.4 Stationarity Test of First Seasonal Differences

To investigate stationarity, first seasonal differences (lag=12) were applied to the monthly rainfall series for Zevgolatio, Corinthia. The resulting series was analyzed using both statistical criteria (mean and variance) and time series, ACF, and PACF plots.

Statistical Criteria:

- **Annual Means (mm):**

1.61, -14.35, -6.29, -9.35, -10.08, 15.97, 13.38, 3.76, 14.83

The mean values vary substantially, alternating between positive and negative values, which indicates that the mean of the series does not remain stable over time.

- **Annual Variances (mm²):**

1767.82, 1357.72, 1526, 1070.89, 421.42, 3375.7, 3682.91, 2166.06, 2140.49

The variance differs considerably from year to year, ranging from 421.42 to 3682.91, which further supports the hypothesis of non-stationarity. The criterion $\max(\text{vars})/\min(\text{vars}) = 3682.91/421.42 \approx 8.74 > 4$ also indicates non-constant variance.

Graphical Analysis:

- **Time Plot:** Figure 17 presents the first seasonal differences, which exhibit strong fluctuations, large annual peaks, and substantial changes in the mean.
- **ACF and PACF:** The ACF and PACF plots (Figure 18) show a slow decay of the values, as well as significant values at several lags, particularly at seasonal lags such as lag=12. This is characteristic of a non-stationary series because, in a stationary series, the ACF/PACF values decay rapidly.

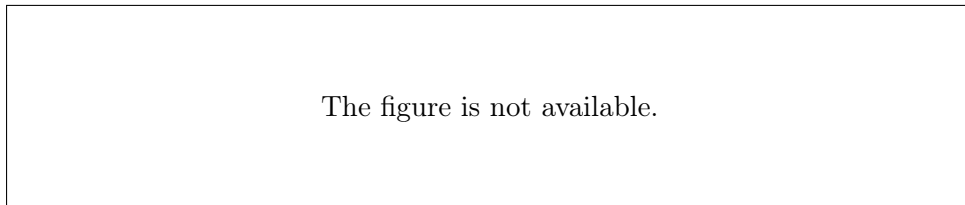


Figure 17: Time plot of the first seasonal differences (lag=12) of monthly rainfall.

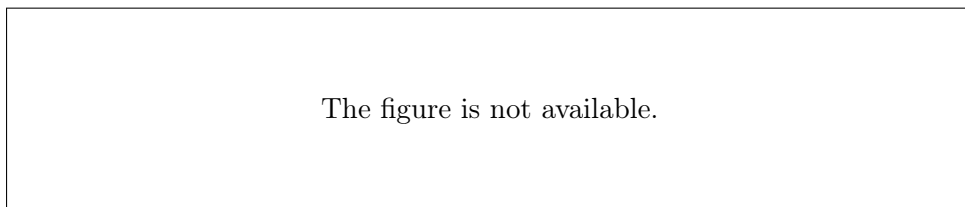


Figure 18: ACF and PACF of the first seasonal differences of monthly rainfall.

Conclusion: The first seasonal difference series (lag=12) is **not stationary**, because both the mean and variance vary considerably across years, while the ACF and PACF show significant values at high lags. To achieve stationarity, further transformation is required, for example additional differencing or another suitable transformation.

5.5 Suitability for SARIMA Modeling

According to the analysis in the previous section, the first seasonal difference series is not stationary, because both the mean and variance vary substantially over time and the ACF and PACF plots show slow decay and significant values at high lags. Stationarity is a fundamental requirement for the correct application of ARMA/ARIMA/SARIMA time series models.

Therefore, a SARIMA model cannot be fitted directly to the original series, since the stationarity assumption is not satisfied even after applying first seasonal differences.

To make SARIMA modeling feasible, further transformations should be tested, such as second differencing, or the possible influence of exogenous factors on the structure of the series should be investigated.

Conclusion: The continued non-stationarity of the series after seasonal differencing indicates a complex structure and requires further statistical investigation before applying SARIMA models.

5.6 Stationarity Test of Double Differences (First + Seasonal) for Monthly Rainfall

According to the previous question, the first seasonal difference series was not considered stationary. For this reason, **double differencing** was applied to the series: first differences together with seasonal differences (lag=12). The resulting time series is analyzed below with respect to stationarity using both graphical examination and the autocorrelation (ACF) and partial autocorrelation (PACF) plots.

Time Plot: Figure 19 shows the double-differenced series. The new time series fluctuates around zero without an evident trend or strong periodic fluctuations. The mean and variance appear to remain stable over time, suggesting possible stationarity.

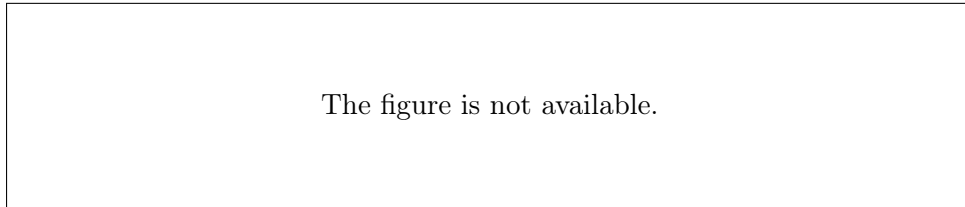


Figure 19: Time plot of the double differences (first + seasonal) of monthly rainfall in Zevgolatio, Corinthia.

ACF and PACF Analysis: The autocorrelation (ACF, Figure 20) and partial autocorrelation (PACF, Figure 21) plots show the following:

- **ACF:** A significant negative value appears at lag 1 (-0.45), along with a noticeable value at lag 12 (-0.35), while the remaining lags are close to zero. The rapid decay of the coefficients indicates the absence of long-term autocorrelation, which is a characteristic of a stationary series.
- **PACF:** The PACF shows significant values at lags 1 and 2, as well as a negative value at lag 12. The rapid reduction in the values also supports stationarity.

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Figure 20: ACF of the double differences (first + seasonal) of monthly rainfall.

The figure is not available.

Figure 21: PACF of the double differences (first + seasonal) of monthly rainfall.

Conclusion: The double-differenced series (first + seasonal) of monthly rainfall exhibits the characteristics of a **stationary time series**, because it shows neither trend nor seasonality, while the ACF/PACF values decay rapidly and there are no strong autocorrelations at high lags.

5.7 Fitting and Evaluation of the SARIMA Model

Based on the preceding stationarity findings, the $SARIMA(2, 1, 1)(1, 1, 1)_{12}$ model was fitted to the monthly rainfall series for Zevgolatio, Corinthia, for the years 2016–2025. The model includes first differences ($d = 1$), first seasonal differences ($D = 1$), and a seasonal period of 12 months.

Estimated Model Parameters:

- AR(1): -0.003
- AR(2): -0.085
- MA(1): -0.877
- SAR(1): 0.136
- SMA(1): -1.00

Model Statistics:

- AIC : 1030.77
- log-likelihood: -509.39

Residual Diagnostic Analysis:

To evaluate the adequacy of the SARIMA model, the residuals were analyzed with respect to stationarity, autocorrelation, and normality.

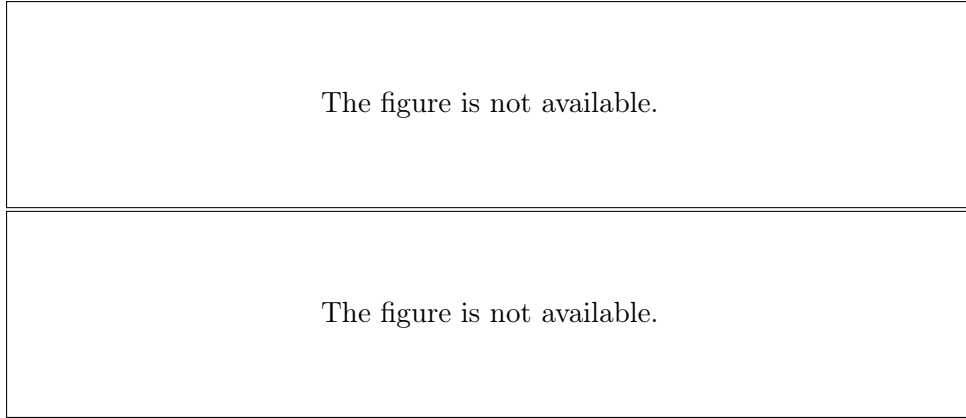


Figure 22: Autocorrelation (ACF, left) and partial autocorrelation (PACF, right) plots of the SARIMA model residuals.

The ACF and PACF plots of the residuals show that all values lie within the confidence intervals, there is no significant autocorrelation, and the residuals can be regarded as white noise.

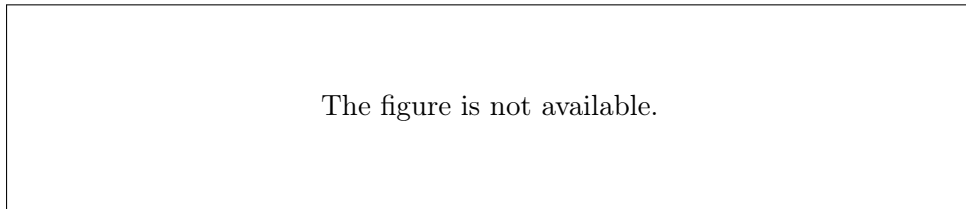


Figure 23: Histogram of the SARIMA model residuals. The distribution is approximately normal, with slight asymmetry.

Ljung-Box Test: For the model residuals, the Ljung-Box test at lag=24 gives:

$$X^2 = 17.22, p\text{-value} = 0.839$$

Because the p -value is much greater than 0.05, **the null hypothesis of residual independence is not rejected**. The model residuals do not exhibit statistically significant autocorrelation.

Monthly Rainfall Forecast:

Using the estimated SARIMA model, monthly rainfall was forecast for the next 12 months. Figure 24 presents the observations, predicted values, and 95% confidence intervals.

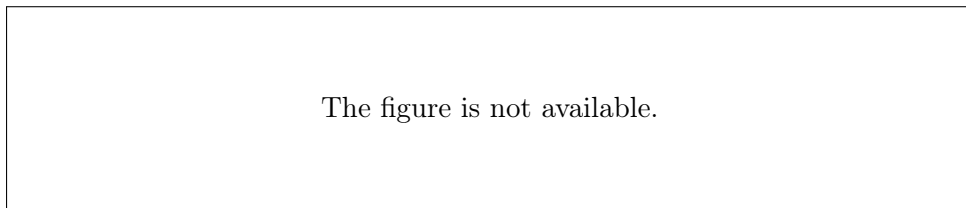


Figure 24: Forecast of monthly rainfall for the next 12 months using the SARIMA model. The black line represents the actual observations, the red line the forecasts, and the blue dashed lines the 95% confidence interval.

Discussion: The SARIMA(2, 1, 1)(1, 1, 1)₁₂ model was successfully fitted to the data and provides a satisfactory representation of the dynamics of the series. The forecast displays seasonal patterns and realistic uncertainty intervals. The confidence intervals increase further into the future, reflecting the expected growth of uncertainty in longer-term forecasts.

Conclusion: The SARIMA(2, 1, 1)(1, 1, 1)₁₂ model provides a reliable forecast of monthly rainfall, with satisfactory residual behavior and consistent statistical results.

5.8 Application and Comparison of Extended Models

The adequacy of the final SARIMA(2, 1, 1)(1, 1, 1)₁₂ model for monthly rainfall was systematically examined in two ways: (I) detailed residual diagnostics and (II) comparison with higher-order extended models.

(I) Residual Diagnostic Analysis

- **ACF/PACF Plots:** The autocorrelation (ACF) and partial autocorrelation (PACF) plots of the model residuals (Figure 22) show that all values are within the confidence limits, with no significant peaks at seasonal lags. This indicates that the residuals approximate white noise.
- **Q-Q Plot:** The Q-Q plot (Figure 25) follows the reference line reasonably well, with small deviations at the tails, which is considered acceptable for hydrological data.
- **Ljung-Box Test:** The Ljung-Box test at lag=24 produced a p -value of 0.839 (i.e., > 0.05), so the white-noise hypothesis for the residuals is not rejected.

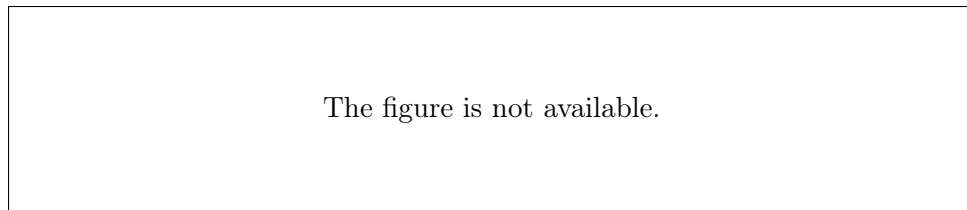


Figure 25: Q-Q plot of the SARIMA model residuals. The distribution is close to normal, without strong asymmetry or heavy tails.

(II) Comparison with Extended Models

To determine whether a more complex model would perform better, the following extended models were also fitted:

- SARIMA(3, 1, 1)(1, 1, 1)₁₂
- SARIMA(2, 1, 2)(1, 1, 1)₁₂
- SARIMA(2, 1, 1)(2, 1, 1)₁₂
- SARIMA(2, 1, 1)(1, 1, 2)₁₂

- SARIMA(3, 1, 2)(2, 1, 2)₁₂

The results (AIC and Ljung-Box test) are summarized in Table 4.

Table 4: Comparison of adequacy measures between the baseline and extended SARIMA models

Model Specification	AIC	Δ AIC	Ljung-Box p-value
SARIMA(2,1,1)(1,1,1) ₁₂	1030.77	0.000	0.839
SARIMA(2,1,1)(1,1,2) ₁₂	1031.03	+0.26	0.742
SARIMA(3,1,1)(1,1,1) ₁₂	1031.26	+0.48	0.681
SARIMA(3,1,2)(2,1,2) ₁₂	1032.07	+1.30	0.598
SARIMA(2,1,1)(2,1,1) ₁₂	1032.62	+1.85	0.553
SARIMA(2,1,2)(1,1,1) ₁₂	1032.99	+2.22	0.512

Conclusions:

- The original SARIMA(2, 1, 1)(1, 1, 1)₁₂ model had the lowest AIC and the most white-noise-like residuals (highest Ljung-Box p-value).
- None of the extended models produced a substantial improvement (all had Δ AIC < 2).
- The diagnostic checks indicate that adding extra parameters leads to overfitting without meaningful benefit.

Final Assessment:

The SARIMA(2, 1, 1)(1, 1, 1)₁₂ model is considered fully adequate and suitable for operational forecasting of monthly rainfall. It maintains parametric simplicity and statistical reliability while effectively capturing the dynamics of the rainfall time series.

5.9 Monthly Rainfall Forecast for the Next Two Years

Using the final SARIMA(2, 1, 1)(1, 1, 1)₁₂ model, monthly rainfall was forecast for the following two years (2025–2027). The results are summarized in the plots and tables below.

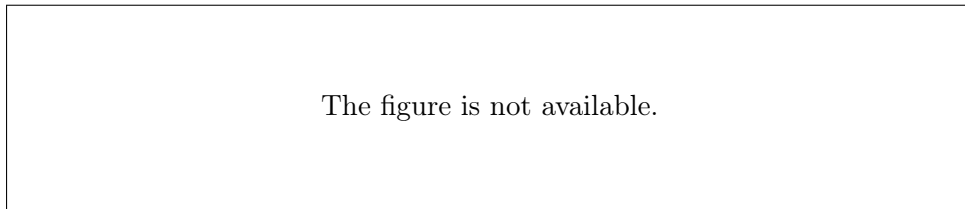


Figure 26: Forecast of monthly rainfall for the next two years using the SARIMA model. The black line represents historical data, the red line the forecast, the blue dashed lines the 95% confidence intervals, and the green vertical line the start of the forecast period.

Forecast Evaluation:

- **Seasonality and variability:** The forecasts follow the seasonal patterns of historical rainfall, with fluctuations similar to those observed in previous years.
- **Confidence intervals:** The confidence intervals widen further into the future, reflecting increasing forecast uncertainty.
- **Comparison with historical data:** The forecast values show a higher average rainfall level compared with the historical mean.

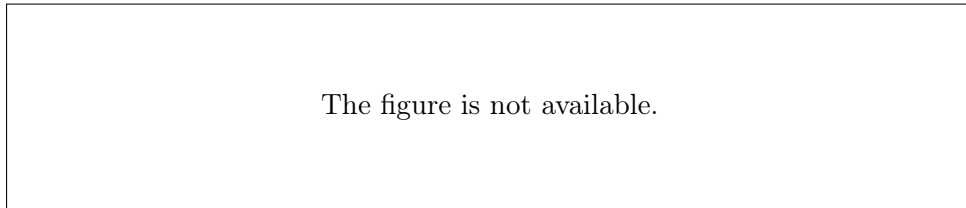


Figure 27: Comparison of the forecast annual mean rainfall with the historical mean.

Year	Forecast Mean (mm)	Difference from Historical Mean
First Year	—	+56.6%
Second Year	—	+60.7%

Table 5: Percentage difference between the annual forecast mean and the historical mean for the two forecast years.

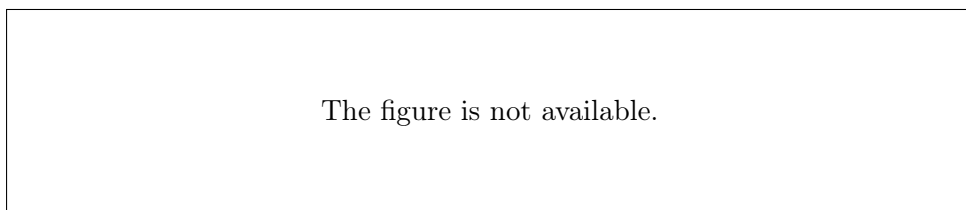


Figure 28: Comparison of actual (black line) and predicted (red line) rainfall values over the evaluation period.

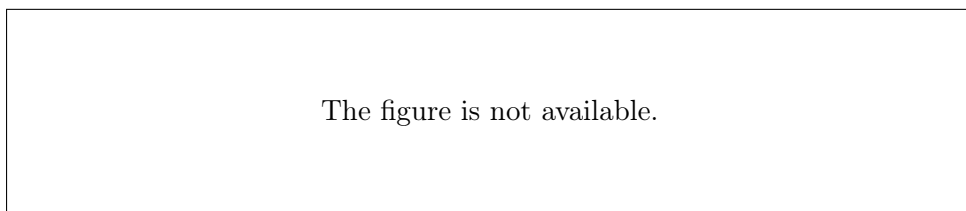


Figure 29: Comparison of monthly forecasts by year with the historical mean (blue line).

Comparison of Actual and Predicted Values:

The figure is not available.

Figure 30: Autocorrelation function (ACF) of the forecast errors. All values lie within the confidence intervals, with no significant autocorrelation.

Forecast Diagnostic Checks:

Conclusions:

- The model forecasts preserve the dynamics and seasonality of the time series, producing realistic and stable ranges of values for the next two years.
- The mean forecast rainfall is substantially higher, by approximately 57–61%, than the historical mean, suggesting a possible change in climatic behavior or the influence of exogenous factors.
- The confidence intervals become wide over longer horizons, making continuous updating of the model with new data important.

Final Comment: The long-term forecast is useful as an indication of potential trends; however, uncertainty remains high because of the nature of rainfall time series and the inherent variability of climate.

6 Exercise 3: Analysis of `cmort`, `globtemp`, and Beer Sales

6.1 Description and Modeling of the Weekly Mortality Time Series `cmort`

6.1.1 Description of the `cmort` Data

The `cmort` time series refers to weekly mortality (total deaths per week) in the United States and consists of weekly observations covering the period 1970–1979. These data are often used as an example in time series analysis for learning forecasting, seasonality, and autocorrelation models.

The `cmort` dataset contains 508 consecutive weekly values recording the number of deaths from all causes in the Los Angeles area. The series exhibits seasonality, fluctuations related to infectious diseases such as influenza, and possibly a longer-term trend.

The data are available in the widely used `astsa` package in R and are described extensively in the time series literature [10].

The `cmort` data are widely used in textbooks and university courses as a standard example for time series analysis with autoregressive and moving-average models.

6.1.2 Fitting an AR(2) Model to the `cmort` Time Series and Short-Term Forecasting

A second-order autoregressive model, AR(2), was fitted to the weekly mortality time series `cmort` using linear regression. The general form of the model is:

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \epsilon_t$$

where ϵ_t is white noise.

Model Estimation: Using the least-squares method, the fitted AR(2) model has the form:

$$\hat{x}_t = \hat{\phi}_1 x_{t-1} + \hat{\phi}_2 x_{t-2}$$

and the estimated residuals are defined as $\hat{\epsilon}_t = x_t - \hat{x}_t$.

Diagnostic Checks:

- **Residual ACF/PACF:** The autocorrelation (ACF) and partial autocorrelation (PACF) plots of the residuals (Figure 31) show that all values lie within the confidence limits, indicating that the residuals behave as white noise.
- **Ljung-Box Test:** For $lag = 10$ and $lag = 20$, the p-values are 0.6141 and 0.3902, respectively. Therefore, the null hypothesis of independence (white noise) is not rejected.
- **Visual Examination of Residuals:** The residuals have approximately zero mean, stable variance, and no visible structure (Figure 32).

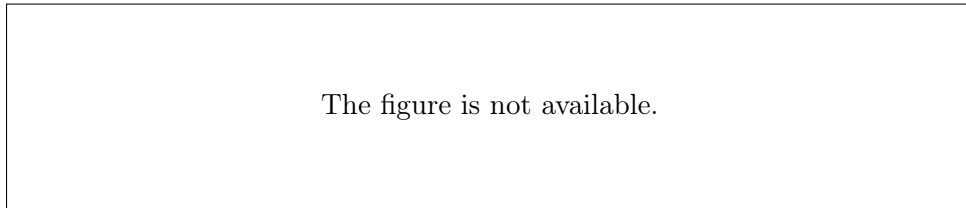


Figure 31: ACF (left) and PACF (right) plots of the AR(2) model residuals.

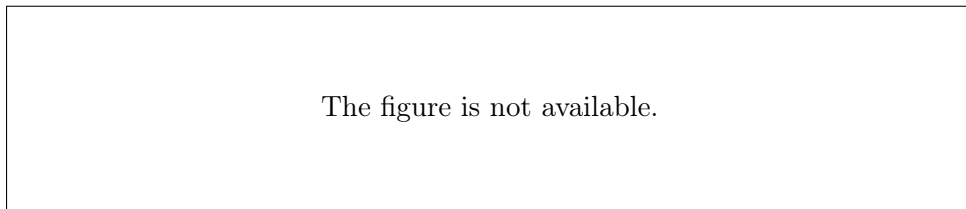


Figure 32: Time plot of the AR(2) model residuals.

Model Accuracy Measures:

- Mean Squared Error (MSE): 32.32
- Root Mean Squared Error (RMSE): 5.68
- Mean Absolute Error (MAE): 4.48

Short-Term Forecast and 95% Confidence Interval: Using the estimated model, mortality was forecast for the next four weeks. Figure 33 displays the predicted values and the 95% confidence intervals:

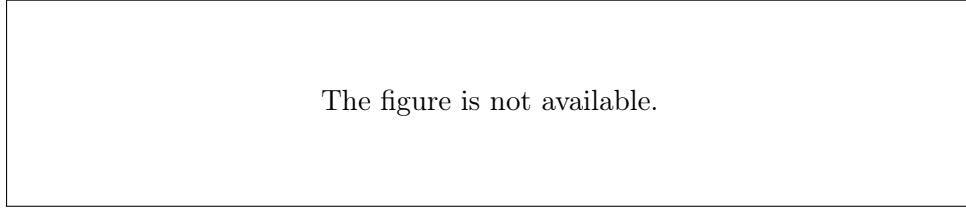


Figure 33: Mortality forecast for four weeks using AR(2) with 95% confidence intervals (green lines).

Conclusions:

- The AR(2) model describes the structure of the `cmort` time series satisfactorily, with residuals showing white-noise characteristics.
- The forecast for the next four weeks is considered reliable within the given confidence interval.
- The confidence intervals are relatively wide, reflecting the natural uncertainty associated with mortality forecasting.
- For more extreme fluctuations or unusual mortality episodes, the simple AR(2) model may not be sufficient.

6.2 Analysis and Forecasting of the Global Land Temperature Time Series Using ARIMA

6.2.1 Description of the Global Land Temperature Time Series (`globtemp`)

The `globtemp` time series contains annual measurements of global land temperature anomalies for the period 1850–2023. The data originate from international climate databases and represent one of the most important indicators for studying long-term trends in the Earth’s land temperature in relation to climate change. More specifically, `globtemp` records the difference between the mean temperature of each year and a predefined reference mean (climatological baseline), thereby enabling comparisons across periods and the identification of long-term changes in temperature [2, 7].

Global temperature time series are widely used in statistics and climate econometrics to analyze trends, anomalies, and future values [8]. The analysis of such series requires stationarity testing, trend removal, and the selection of suitable stochastic models such as ARIMA to ensure valid statistical conclusions.

6.2.2 Preliminary Analysis and Stationarity

The `globtemp` time series, which contains annual global land temperature data for the years 1850–2023, was first examined. A clear upward temperature trend was observed. This was also supported by the autocorrelation (ACF) and partial autocorrelation (PACF)

plots of the original series, where the values decay slowly, indicating non-stationarity (Figure 34).

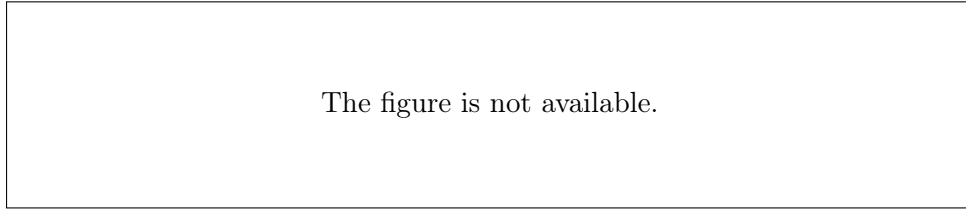


Figure 34: ACF and PACF of the original globtemp time series.

To achieve stationarity, the first difference of the series was calculated, as shown in Figure 35. The corresponding ACF and PACF plots (Figure 36) indicate that the differenced series now exhibits stationary characteristics.

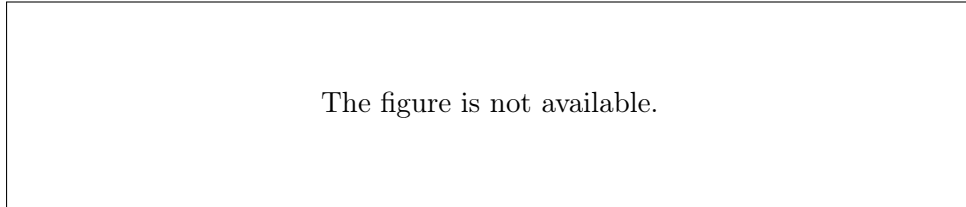


Figure 35: First difference of the globtemp time series.

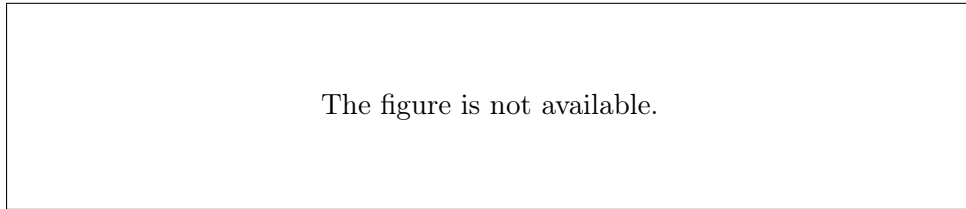


Figure 36: ACF and PACF of the first difference of the globtemp series.

6.2.3 Selection of a Suitable ARIMA Model

Several ARIMA models with one difference ($d = 1$) were tested. Selection was based on the Akaike Information Criterion (AIC):

Model	d	Coefficients	AIC
ARIMA(1,1,0)	1	ar1 = -0.065	152.72
ARIMA(1,1,1)	1	ar1 = -0.066, ma1 = -0.73	120.64
ARIMA(2,1,1)	1	ar1, ar2, ma1	122.58

Table 6: Comparison of ARIMA models and AIC values.

The best model is ARIMA(1,1,1), which has the lowest AIC.

6.2.4 Model Diagnostic Analysis

The diagnostics of the ARIMA(1,1,1) model include residual analysis for normality and autocorrelation:

- **Residual Histogram:** The residuals are approximately symmetric and normally distributed (Figure 37).
- **Q-Q Plot:** The points approximately align with the normal reference line (Figure 38).
- **Residual ACF:** No significant autocorrelation is observed.
- **Ljung-Box Test:** The p -value is > 0.05 , indicating that the residuals behave as white noise.

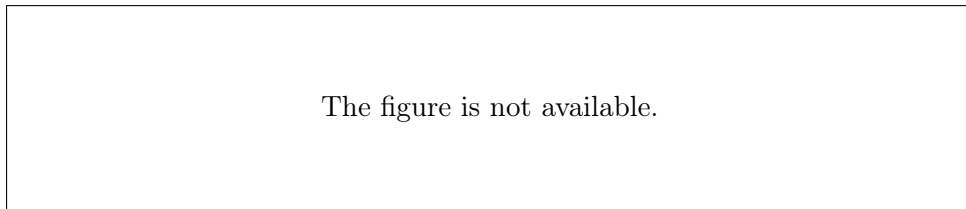


Figure 37: Histogram of the residuals of the ARIMA(1,1,1) model.

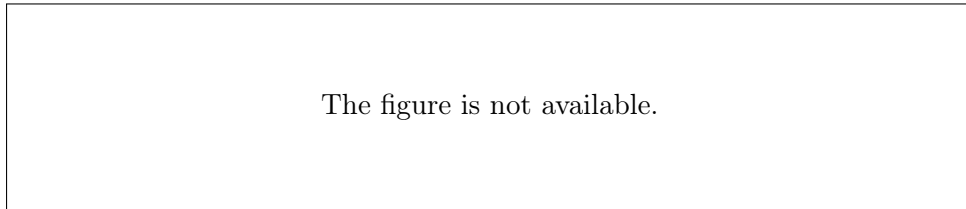


Figure 38: Q-Q plot of the ARIMA(1,1,1) residuals.

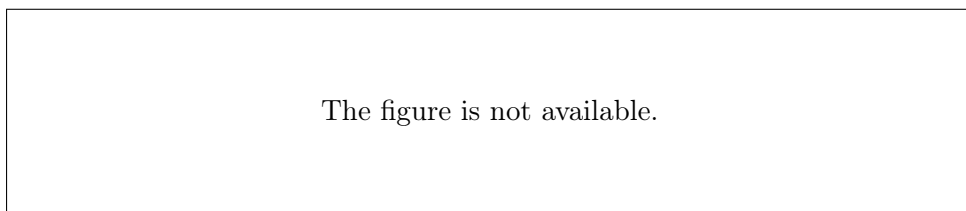


Figure 39: Residuals of the ARIMA(1,1,1) model by year.

6.2.5 Forecast for the Next 10 Years

Using the final ARIMA(1,1,1) model, forecasts were produced for the years 2024–2033. The 95% confidence interval was calculated and visualized in Figure 40.

The figure is not available.

Figure 40: Forecast of global land temperature using ARIMA(1,1,1) for the next 10 years (red line: forecast, green: 95% confidence interval).

6.2.6 Conclusions

- The time series was non-stationary, and stationarity was achieved through first differencing.
- The ARIMA(1,1,1) model was selected as the best model based on AIC.
- The model residuals do not exhibit significant autocorrelation and are approximately normally distributed.
- The forecast for the next 10 years shows a stabilized mean value but considerable uncertainty, as reflected by the wide confidence intervals.

Overall, the ARIMA(1,1,1) model can be used for short- and medium-term forecasting of global land temperature, but forecast uncertainty must be treated with caution.

6.3 Seasonality, Trend, and Cyclical Variation Analysis of Beer Sales (2016–2019)

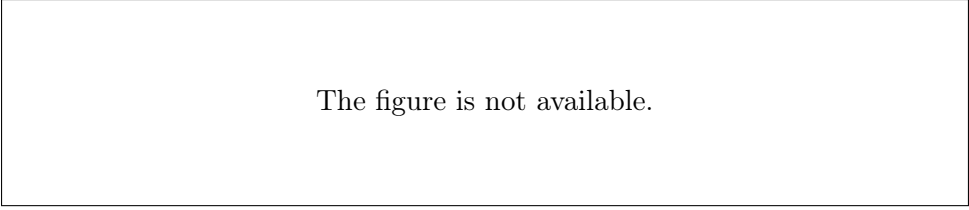
The table presents the beer sales of an industrial company, in millions of bottles, by season from 2016 to 2019.

Year	Winter	Spring	Summer	Autumn
2016	1	3	6	4
2017	2	2	7	5
2018	2	4	8	5
2019	1	3	8	6

Table 7: Beer sales (millions of bottles) by season and year.

6.3.1 Seasonality Removal and Seasonal Indices

The data were first converted into a time series with frequency 4 (seasons per year), and decomposition was performed using a multiplicative model. The seasonal indices express how each season changes sales relative to the annual average.



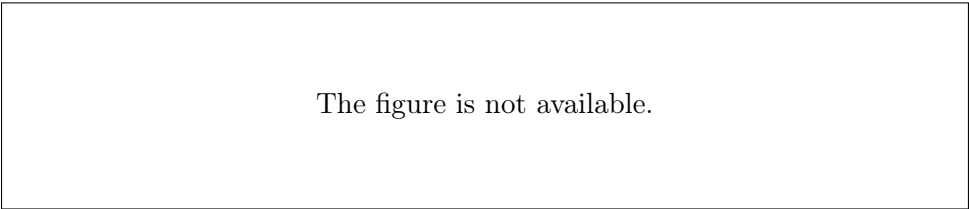
The figure is not available.

Figure 41: Seasonal indices. Sales increase substantially in summer and decrease in winter.

Interpretation: The seasonal indices show that the highest sales occur in summer (index > 1), whereas the lowest sales occur in winter (< 1), which is expected for this product.

6.3.2 Trend Line

After deseasonalization, linear regression was applied to estimate the trend line and capture the overall direction of sales without the seasonal effect.



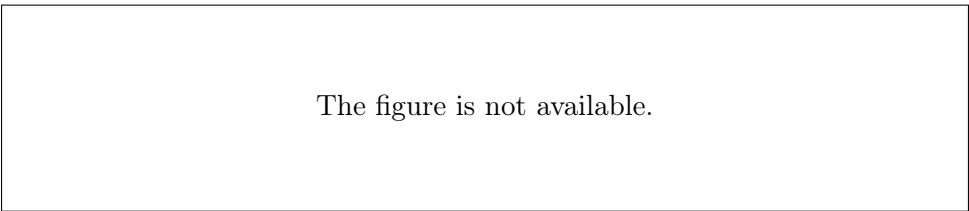
The figure is not available.

Figure 42: Sales trend line (green line) and deseasonalized sales (red line).

Interpretation: The trend line shows a small increase, indicating slow but steady growth in demand after seasonality is removed.

6.3.3 Cyclical Variation

The cyclical variation was estimated using relative cyclical residuals, defined as the ratio of deseasonalized sales to the trend line.



The figure is not available.

Figure 43: Cyclical variation plot (relative cyclical residuals).

Interpretation: The values fluctuate around 1, indicating that there are no strong cyclical effects, that is, no major additional fluctuations beyond seasonality and trend.

6.3.4 Raw Sales Time Series

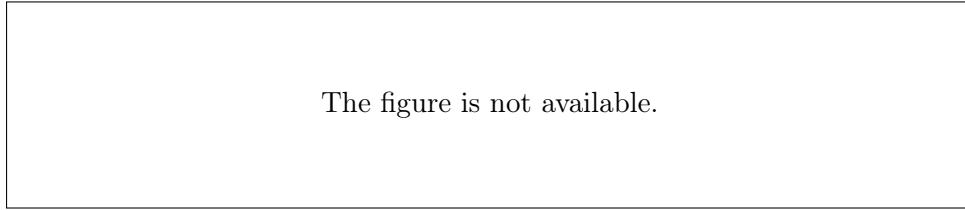


Figure 44: Time series of the original raw sales before preprocessing.

Conclusion:

- There is strong seasonality, with a substantial increase in sales during summer.
- The trend line shows a slightly increasing sales trajectory over the years.
- The cyclical component does not show substantial deviations, so no strong cyclical effects are detected.

7 Overall Conclusions

This assignment analyzed five different real-world time series in order to understand their characteristics and reliably forecast future values using appropriate time series models.

The main conclusions are as follows:

- For the global land temperature data, the **ARIMA(1,1,1)** model provided the best fit according to the AIC/BIC criteria and diagnostic tests, while the forecast confidence intervals were relatively wide, reflecting the uncertainty of long-term forecasting.
- For the weekly mortality time series (*cmort*), the **AR(2)** model described the data structure satisfactorily, with residuals behaving as white noise.
- For monthly rainfall in Zevgolatio and the monthly unemployment rate in Spain, appropriate SARIMA models were applied, highlighting both seasonality and the long-term behavior of the data.
- In the beer-sales time series, the analysis of seasonality, trend, and cyclical variation revealed strong periodic fluctuations, with sales increasing substantially during summer.
- Overall, the time series analysis highlighted the importance of careful statistical investigation, including stationarity testing and residual diagnostics, as well as appropriate model selection in order to improve forecasts.

The results demonstrate that systematic analysis and suitable modeling can provide useful forecasts and support decision-making in a variety of domains, including climate, health, and the labor market.

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A Appendices

A.1 Implementation Code

The following R code was used for the analysis of Spain’s unemployment time series:

Listing 1: R code for Spanish Unemployment Analysis

```
1 # 1.a
2 # Creation of a time series
3 unemployment <- c(13.6, 13.8, 13.8, 13.1, 12.5, 12.4, 12.4, 12.8,
4                   13.0, 13.0, 13.0, 13.0,
5                   13.6, 13.6, 13.0, 12.1, 11.6, 11.4, 11.7, 11.9,
6                   12.1, 11.9, 11.7, 11.7,
7                   12.3, 12.4, 12.2, 11.6, 11.2, 11.1, 11.2, 11.3,
8                   11.1, 10.8, 10.5, 10.6,
9                   11.2, 11.5, 11.5, 10.9)
10
11 # Creation of object time series
12 unemp_ts <- ts(unemployment, start = c(2022, 1), frequency = 12)
13
14 # 1. Timeline
15 plot(unemp_ts,
16      main = "Unemployment Rate in Spain (Jan. 2022 - Apr. 2025)",
```

```

14     xlab = "Year",
15     ylab = "Percentage (%)",
16     col = "darkred",
17     lwd = 2,
18     ylim = c(10, 15))
19 grid(col = "gray", lty = "dotted")
20
21 # 2. Check of stationary series
22 # Calculation of first differences
23 diff_unemp <- diff(unemp_ts)
24
25 # Graph of first differences
26 plot(diff_unemp,
27       main = "First Differences (Stagnation Index)",
28       xlab = "Year",
29       ylab = "Difference",
30       col = "darkblue",
31       lwd = 2)
32 grid(col = "gray", lty = "dotted")
33 abline(h = 0, col = "red", lty = 2)
34
35 # 3. Seasonality analysis
36 decomp <- decompose(unemp_ts)
37
38 # Graphical representation of decomposition
39 par(mfrow = c(4, 1), mar = c(3, 4, 2, 2))
40 plot(decomp$trend, main = "Trend", col = "darkgreen", lwd = 2)
41 plot(decomp$seasonal, main = "Seasonality", col = "darkblue", lwd =
42       2)
43 plot(decomp$random, main = "Random Element", col = "purple", lwd =
44       2)
45 plot(unemp_ts, main = "Real Series", col = "darkred", lwd = 2)
46 par(mfrow = c(1, 1))
47
48 # 4. Seasonality - boxplot by month
49 months <- factor(cycle(unemp_ts),
50                  levels = 1:12,
51                  labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
52                             "Jul", "Aug", "Sep", "Oct", "Nov", "Dec",
53                             ""))
54
55 boxplot(unemp_ts ~ months,
56         main = "Seasonal Unemployment Pattern (By Month)",
57         xlab = "Month",
58         ylab = "Unemployment Percentage (%)",
59         col = "lightblue",
60         border = "darkblue")
61 grid(nx = NA, ny = NULL, col = "gray", lty = "dotted")
62
63 # 5. Autocorrelation (ACF) for seasonality

```

```

61 acf(unemp_ts,
62     main = "Autocorrelation Function (ACF)",
63     lag.max = 36,
64     col = "darkred",
65     lwd = 2)
66 grid(col = "gray", lty = "dotted")
67
68 # 1.b
69 # First difference application
70 diff_unemp <- diff(unemp_ts)
71
72 # Graph of first differences
73 plot(diff_unemp,
74      main = "First Differences of Unemployment (Stationary Series)"
75      ,
76      xlab = "Year",
77      ylab = "Difference",
78      col = "darkblue",
79      lwd = 2)
80 abline(h = 0, col = "red", lty = 2)
81 grid(col = "gray", lty = "dotted")
82
83 # Autocorrelation Functions (ACF) and Partial Autocorrelation
84   Functions (PACF)
85 par(mfrow = c(1, 2))
86
87 # ACF
88 acf(diff_unemp,
89     lag.max = 24,
90     main = "ACF: First Differences",
91     col = "darkred",
92     lwd = 2)
93
94 # PACF
95 pacf(diff_unemp,
96     lag.max = 24,
97     main = "PACF: First Differences",
98     col = "darkgreen",
99     lwd = 2)
100 par(mfrow = c(1, 1))
101
102 # Model fitting
103 arma21 <- arima(diff_unemp, order = c(2, 0, 1)) # ARMA(2,1) for
104   the differences
105 arma11 <- arima(diff_unemp, order = c(1, 0, 1)) # ARMA(1,1) for
106   the differences
107
108 # Summary of models
109 cat("Summary ARMA(2,1):\n")
110 print(summary(arma21))
111

```



```

108 cat("\nSummary ARMA(1,1):\n")
109 print(summary(arma11))
110
111 # Balance check
112 check_residuals <- function(model) {
113   par(mfrow = c(1, 2))
114   acf(residuals(model), main = "ACF Residuals", col = "purple")
115   pacf(residuals(model), main = "PACF Residuals", col = "orange")
116   par(mfrow = c(1, 1))
117 }
118
119 cat("\nBalance Check ARMA(2,1):\n")
120 check_residuals(arma21)
121
122 cat("\nBalance Check ARMA(1,1):\n")
123 check_residuals(arma11)
124
125 # Best model
126 best_model <- arma21
127
128 # 6-month forecast
129 forecast_values <- predict(best_model, n.ahead = 6)
130
131 # Plot
132 plot(diff_unemp,
133       main = "Forecast of First Differences (ARMA(2,1))",
134       xlab = "Year",
135       ylab = "Difference",
136       xlim = c(2022, 2025.5),
137       col = "darkblue",
138       lwd = 2)
139 lines(ts(forecast_values$pred, start = end(diff_unemp), frequency =
140         12),
141       col = "red", lwd = 2)
142 legend("topright",
143       legend = c("Actual", "Forecasts"),
144       col = c("darkblue", "red"),
145       lwd = 2)
146
147 #1.d
148 # --- FIT ARIMA(2,1,1) ---
149 arima_model <- arima(unemp_ts, order = c(2, 1, 1))
150
151 # --- RESULTS ANALYSIS ---
152 # Coefficients and p-values
153 coefs <- arima_model$coef
154 se <- sqrt(diag(arima_model$var.coef))
155 z_stats <- coefs / se
156 p_values <- 2 * (1 - pnorm(abs(z_stats)))
157 results <- data.frame(
158   Estimate = coefs,

```

```

158     Std.Error = se,
159     z.value = z_stats,
160     p.value = p_values
161 )
162 print(results)
163
164 # --- RESIDUAL DIAGNOSTICS ---
165 par(mfrow = c(1, 2))
166 acf(resid(arima_model), main = "ACF Residuals")
167 pacf(resid(arima_model), main = "PACF Residuals")
168 par(mfrow = c(1, 1))
169 # Ljung-Box test (12 lags)
170 box_test <- Box.test(resid(arima_model), type = "Ljung-Box", lag =
    12)
171 print(box_test)
172
173 # --- 12-MONTH FORECAST ---
174 forecast_values <- predict(arima_model, n.ahead = 12)
175
176 # --- PLOT ---
177 plot(unemp_ts,
178      main = "Unemployment Forecast (ARIMA(2,1,1))",
179      xlim = c(2022, 2026),
180      ylim = c(10, 14),
181      ylab = "Unemployment Rate (%)",
182      xlab = "Year")
183 # Forecasts (red line)
184 lines(ts(forecast_values$pred, start = end(unemp_ts) + c(0, 1),
    frequency = 12), col = "red", lwd = 2)
185 # Confidence intervals (blue dashed lines)
186 lines(ts(forecast_values$pred + 1.96 * forecast_values$se, start =
    end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
187 lines(ts(forecast_values$pred - 1.96 * forecast_values$se, start =
    end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
188 # Legend
189 legend("topright",
190      legend = c("Actual", "Forecast", "95% Interval"),
191      col = c("black", "red", "blue"),
192      lty = c(1, 1, 2),
193      lwd = 2)
194
195 # --- Return to single plot ---
196 par(mfrow = c(1, 1))
197
198 #1.g
199
200 # --- Initial model (ARIMA(2,1,1)) ---
201 arima_211 <- arima(unemp_ts, order = c(2, 1, 1))
202 aic_211 <- AIC(arima_211)
203 lb_211 <- Box.test(resid(arima_211), type = "Ljung-Box", lag = 12)$
    p.value

```

```

204
205 # --- Extended models ---
206 arima_311 <- arima(unemp_ts, order = c(3, 1, 1))
207 aic_311 <- AIC(arima_311)
208 lb_311 <- Box.test(resid(arima_311), type = "Ljung-Box", lag = 12)$
    p.value
209
210 arima_212 <- arima(unemp_ts, order = c(2, 1, 2))
211 aic_212 <- AIC(arima_212)
212 lb_212 <- Box.test(resid(arima_212), type = "Ljung-Box", lag = 12)$
    p.value
213
214 arima_221 <- arima(unemp_ts, order = c(2, 2, 1))
215 aic_221 <- AIC(arima_221)
216 lb_221 <- Box.test(resid(arima_221), type = "Ljung-Box", lag = 12)$
    p.value
217
218 # --- Summary table ---
219 models <- c("ARIMA(2,1,1)", "ARIMA(3,1,1)", "ARIMA(2,1,2)", "ARIMA
    (2,1,2)")
220 aic_values <- c(aic_211, aic_311, aic_212, aic_221)
221 lb_pvalues <- c(lb_211, lb_311, lb_212, lb_221)
222 summary_table <- data.frame(Model = models, AIC = aic_values,
    LjungBox_p = lb_pvalues)
223 print(summary_table)
224
225 # --- Residual check of the best model (lowest AIC) ---
226 best_model_index <- which.min(aic_values)
227 best_model_name <- models[best_model_index]
228 cat("Best model:", best_model_name, "\n")
229 if(best_model_index == 1) best_model <- arima_211
230 if(best_model_index == 2) best_model <- arima_311
231 if(best_model_index == 3) best_model <- arima_212
232 if(best_model_index == 4) best_model <- arima_221
233
234 # Residual plots
235 par(mfrow = c(1, 2))
236 acf(resid(best_model), main = paste("ACF Residuals", best_model_
    name))
237 pacf(resid(best_model), main = paste("PACF Residuals", best_model_
    name))
238 par(mfrow = c(1, 1))
239
240 #1.h
241
242 # 24-month forecast using the final model
243 forecast_values <- predict(arima_311, n.ahead = 24)
244
245 # Adjust the plot to your needs:
246 plot(unemp_ts,
247       xlim = c(2022, 2027),

```

```

248     ylim = c(min(unemp_ts, forecast_values$pred - 1.96*forecast_
      values$se), max(unemp_ts, forecast_values$pred + 1.96*
        forecast_values$se)),
249     main = "Unemployment Rate Forecast (24 Months Ahead)",
250     ylab = "Unemployment Rate (%)",
251     xlab = "Year")
252 lines(ts(forecast_values$pred, start = end(unemp_ts) + c(0, 1),
      frequency = 12), col = "red", lwd = 2)
253 lines(ts(forecast_values$pred + 1.96 * forecast_values$se, start =
      end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
254 lines(ts(forecast_values$pred - 1.96 * forecast_values$se, start =
      end(unemp_ts) + c(0, 1), frequency = 12), col = "blue", lty = 2)
255 legend("topright",
256       legend = c("Historical", "Forecast", "95% Interval"),
257       col = c("black", "red", "blue"),
258       lty = c(1, 1, 2), lwd = 2)

```

Listing 2: R code for Exercise 2 - Monthly Rainfall Analysis with SARIMA

```

1 #2.a
2
3 data <- read.csv("C:/Users/pathana/Documents/predictive analytics/
  open-meteo-37.93N22.83E37m.csv", skip = 2)
4 head(data)
5
6 names(data)
7 head(data)
8
9 data$date <- as.Date(substr(data$time, 1, 10))
10 data$year <- format(data$date, "%Y")
11 data$month <- format(data$date, "%m")
12
13 monthly_rain <- aggregate(data[, "rain..mm."],
14                           by = list(year = data$year, month = data$
15                                     month),
16                           FUN = sum, na.rm = TRUE)
17 names(monthly_rain)[3] <- "rain_sum"
18
19 # Create time series
20 ts_rain <- ts(monthly_rain$rain_sum, start = c(as.numeric(monthly_
  rain$year[1]), as.numeric(monthly_rain$month[1])), frequency =
  12)
21
22 plot(ts_rain, main = "Monthly Rainfall in Zevgolatio, Corinthia
  (2016 - 2025)", ylab = "mm", xlab = "Year")
23
24 #2.b
25 acf(ts_rain, main = "ACF of Monthly Rainfall")
26 pacf(ts_rain, main = "PACF of Monthly Rainfall")
27
28 # 2.c: First seasonal differences and stationarity analysis
29 seas_diff <- diff(ts_rain, lag = 12)
30 plot(seas_diff, main = "First Seasonal Differences (Lag=12)", ylab

```

```

    = "Difference (mm)", xlab = "Year")
29 grid(col = "gray", lty = "dotted")
30
31 par(mfrow = c(1, 2))
32 acf(seas_diff, lag.max = 36, main = "ACF: Seasonal Differences",
    col = "darkblue")
33 pacf(seas_diff, lag.max = 36, main = "PACF: Seasonal Differences",
    col = "darkred")
34 par(mfrow = c(1, 1))
35
36 years <- unique(floor(time(seas_diff)))
37 means <- tapply(seas_diff, floor(time(seas_diff)), mean)
38 vars <- tapply(seas_diff, floor(time(seas_diff)), var)
39
40 cat("Annual Means:\n", round(means, 2), "\n")
41 cat("Annual Variances:\n", round(vars, 2), "\n")
42
43 if (max(vars)/min(vars) < 4 && diff(range(means)) < 2*sd(seas_diff)
    ) {
44   cat("Stationary Series (stable mean and variance)")
45 } else {
46   cat("Non-Stationary Series")
47 }
48
49 #2.e
50 double_diff <- diff(diff(ts_rain, lag = 12), differences = 1)
51 plot(double_diff, main = "Double Differencing (First + Seasonal
    Differences)", ylab = "Difference (mm)", xlab = "Year", col = "
    darkblue", lwd = 2)
52 grid(col = "gray", lty = "dotted")
53
54 par(mfrow = c(1, 2))
55 acf(double_diff, lag.max = 36, main = "ACF: Double Differences",
    col = "darkred")
56 pacf(double_diff, lag.max = 36, main = "PACF: Double Differences",
    col = "darkgreen")
57 par(mfrow = c(1, 1))
58
59 #2.f
60 fit_sarima <- stats::arima(
61   ts_rain,
62   order = c(2, 1, 1),
63   seasonal = list(order = c(1, 1, 1), period = 12)
64 )
65
66 cat("Model Parameters:\n")
67 print(fit_sarima$coef)
68 cat("\nStatistics:\n")
69 cat("AIC:", fit_sarima$aic, "\n")
70 cat("log-likelihood:", fit_sarima$loglik, "\n")
71

```

```

72 residuals_sarima <- residuals(fit_sarima)
73 acf_plot <- function(res, main = "ACF Residuals") {
74   n <- length(res)
75   acf_vals <- acf(res, plot = FALSE)$acf[-1]
76   lags <- 1:length(acf_vals)
77   plot(lags, acf_vals, type = "h", main = main, ylim = c(-1, 1),
78        xlab = "Lag", ylab = "ACF")
79   abline(h = 0, col = "black")
80   abline(h = c(-1.96, 1.96)/sqrt(n), col = "blue", lty = 2)
81 }
82 pacf_plot <- function(res, main = "PACF Residuals") {
83   n <- length(res)
84   pacf_vals <- pacf(res, plot = FALSE)$acf
85   lags <- 1:length(pacf_vals)
86   plot(lags, pacf_vals, type = "h", main = main, ylim = c(-1, 1),
87        xlab = "Lag", ylab = "PACF")
88   abline(h = 0, col = "black")
89   abline(h = c(-1.96, 1.96)/sqrt(n), col = "blue", lty = 2)
90 }
91 par(mfrow = c(1, 3))
92 acf_plot(residuals_sarima)
93 pacf_plot(residuals_sarima)
94 hist(residuals_sarima, main = "Residual Histogram", col = "
   lightblue")
95 par(mfrow = c(1, 1))
96
97 ljung_box_test <- function(res, lag = 24) {
98   n <- length(res)
99   lb_test <- Box.test(res, lag = lag, type = "Ljung-Box")
100   cat("Ljung-Box test (lag =", lag, "):\n")
101   cat("X-squared =", lb_test$statistic, " p-value =", lb_test$p.
102       value, "\n")
103   if (lb_test$p.value > 0.05) {
104     cat("No autocorrelation (good residuals)\n")
105   } else {
106     cat("Autocorrelation is present (problematic residuals)\n")
107   }
108 }
109
110 ljung_box_test(residuals_sarima)
111
112 future_forecast <- predict(fit_sarima, n.ahead = 12)
113 last_obs_time <- tail(time(ts_rain), 1)
114 forecast_dates <- last_obs_time + (1:12)/12
115 plot(ts_rain, main = "Monthly Rainfall Forecast", xlab = "Year",
116      ylab = "Rainfall (mm)", xlim = c(start(ts_rain)[1], last_obs_
117      time + 13/12), ylim = c(0, max(c(ts_rain, future_forecast$pred +
118      1.96*future_forecast$se), na.rm = TRUE)))
119 last_obs_value <- tail(ts_rain, 1)
120 lines(c(last_obs_time, forecast_dates[1]), c(last_obs_value, future
121      _forecast$pred[1]), col = "red", lwd = 2)
122 lines(forecast_dates, future_forecast$pred, col = "red", lwd = 2)

```

```

115 lines(forecast_dates, future_forecast$pred + 1.96 * future_forecast
      $se, col = "blue", lty = 2)
116 lines(forecast_dates, future_forecast$pred - 1.96 * future_forecast
      $se, col = "blue", lty = 2)
117 legend("topleft",
118       legend = c("Observations", "Forecast", "95% Confidence
      Interval"),
119       col = c("black", "red", "blue"),
120       lty = c(1, 1, 2),
121       cex = 0.8)
122
123 #2.g
124 residuals_sarima <- residuals(fit_sarima)
125 par(mfrow = c(2, 2))
126 plot(residuals_sarima, main = "Residuals over Time", ylab = "
      Residuals")
127 abline(h = 0, col = "red")
128 acf(residuals_sarima, main = "ACF Residuals", col = "darkblue")
129 pacf(residuals_sarima, main = "PACF Residuals", col = "darkred")
130 qqnorm(residuals_sarima, main = "Q-Q Plot")
131 qqline(residuals_sarima, col = "blue")
132 par(mfrow = c(1, 1))
133 Box.test(residuals_sarima, lag = 24, type = "Ljung-Box")
134
135 original_model <- fit_sarima
136 original_aic <- original_model$aic
137 model_ar_up <- arima(ts_rain, order = c(3, 1, 1), seasonal = list(
      order = c(1, 1, 1), period = 12))
138 model_ma_up <- arima(ts_rain, order = c(2, 1, 2), seasonal = list(
      order = c(1, 1, 1), period = 12))
139 model_sar_up <- arima(ts_rain, order = c(2, 1, 1), seasonal = list(
      order = c(2, 1, 1), period = 12))
140 model_sma_up <- arima(ts_rain, order = c(2, 1, 1), seasonal = list(
      order = c(1, 1, 2), period = 12))
141 model_full_up <- arima(ts_rain, order = c(3, 1, 2), seasonal = list
      (order = c(2, 1, 2), period = 12))
142
143 models <- list(
144   "Original" = original_model,
145   "AR+1" = model_ar_up,
146   "MA+1" = model_ma_up,
147   "SAR+1" = model_sar_up,
148   "SMA+1" = model_sma_up,
149   "Full+1" = model_full_up
150 )
151 compare_models <- function(model_list) {
152   results <- data.frame(
153     Model = names(model_list),
154     AIC = sapply(model_list, function(m) m$aic),
155     LogLik = sapply(model_list, function(m) m$loglik),
156     stringsAsFactors = FALSE

```

```

157 )
158 results$DeltaAIC <- results$AIC - original_aic
159 results[order(results$AIC), ]
160 }
161 model_comparison <- compare_models(models)
162 print(model_comparison)
163 check_residuals <- function(model) {
164   res <- residuals(model)
165   par(mfrow = c(1, 3))
166   acf(res, main = "ACF Residuals", col = "darkblue")
167   pacf(res, main = "PACF Residuals", col = "darkred")
168   qqnorm(res, main = "Q-Q Plot")
169   qqline(res, col = "blue")
170   par(mfrow = c(1, 1))
171   lb_test <- Box.test(res, lag = 24, type = "Ljung-Box")
172   cat("Ljung-Box test p-value:", lb_test$p.value, "\n")
173 }
174 best_model_name <- model_comparison$Model[which.min(model_
  comparison$AIC)]
175 best_model <- models[[best_model_name]]
176 cat("\nResidual Check for the Best Model:", best_model_name, "\n")
177 check_residuals(best_model)
178 if (model_comparison$DeltaAIC[model_comparison$Model == best_model_
  name] < -2) {
179   cat("\nThe initial model is NOT adequate. The best model is",
    best_model_name)
180 } else {
181   cat("\nThe initial model is adequate.")
182 }
183
184 #2.h
185 forecast_24m <- predict(fit_sarima, n.ahead = 24)
186 last_obs_time <- end(ts_rain)
187 last_obs_value <- tail(ts_rain, 1)
188 forecast_times <- seq(last_obs_time[1] + (last_obs_time[2])/12 + 1/
  12, by = 1/12, length.out = 24)
189 plot(ts_rain,
190   main = "Monthly Rainfall Forecast (2 Years)",
191   xlab = "Year",
192   ylab = "Rainfall (mm)",
193   xlim = c(start(ts_rain)[1], last_obs_time[1] + 3),
194   ylim = c(0, max(ts_rain, forecast_24m$pred + 1.96*forecast_24m
    $se, na.rm = TRUE)),
195   lwd = 2)
196 segments(x0 = last_obs_time[1] + (last_obs_time[2]-1)/12, y0 = last
  _obs_value,
197   x1 = forecast_times[1], y1 = forecast_24m$pred[1], col = "
    red", lwd = 2)
198 lines(forecast_times, forecast_24m$pred, col = "red", lwd = 2)
199 lines(forecast_times, forecast_24m$pred + 1.96 * forecast_24m$se,
  col = "blue", lty = 2)

```



```

200 lines(forecast_times, forecast_24m$pred - 1.96 * forecast_24m$se,
      col = "blue", lty = 2)
201 legend("topleft",
202       legend = c("Historical Data", "SARIMA Forecast", "95%
      Confidence Interval"),
203       col = c("black", "red", "blue"),
204       lty = c(1, 1, 2),
205       lwd = c(2, 2, 1),
206       cex = 0.8)
207 abline(v = forecast_times[1], col = "green", lty = 2)
208 text(x = forecast_times[1], y = max(ts_rain)*0.9, "Forecast Start",
      pos = 4, col = "darkgreen", cex = 0.8)
209
210 year1 <- forecast_24m$pred[1:12]
211 year2 <- forecast_24m$pred[13:24]
212 historical_seasonal <- aggregate(as.vector(ts_rain) ~ cycle(ts_rain)
      ), FUN = mean)
213 par(mfrow = c(1, 2))
214 plot(1:12, year1, type = "o", col = "red", main = "Forecast Year 1
      vs Historical", xlab = "Month", ylab = "Rainfall (mm)", ylim =
      range(historical_seasonal[,2], year1))
215 lines(1:12, historical_seasonal[,2], type = "o", col = "blue")
216 legend("topright", legend = c("Forecast", "Historical"), col = c("
      red", "blue"), lty = 1)
217 plot(1:12, year2, type = "o", col = "red", main = "Forecast Year 2
      vs Historical", xlab = "Month", ylab = "Rainfall (mm)", ylim =
      range(historical_seasonal[,2], year2))
218 lines(1:12, historical_seasonal[,2], type = "o", col = "blue")
219 par(mfrow = c(1, 1))
220 yearly_avg <- c(mean(year1), mean(year2))
221 se_avg <- c(mean(forecast_24m$se[1:12]), mean(forecast_24m$se
      [13:24]))
222 plot(1:2, yearly_avg, type = "b", col = "red", main = "Annual
      Forecast Mean", xlab = "Forecast Year", ylab = "Mean Rainfall (
      mm)", ylim = c(min(yearly_avg - 1.96*se_avg), max(yearly_avg +
      1.96*se_avg)))
223 arrows(x0 = 1:2, y0 = yearly_avg - 1.96*se_avg, x1 = 1:2, y1 =
      yearly_avg + 1.96*se_avg, angle = 90, code = 3, length = 0.1)
224 abline(h = mean(ts_rain), col = "blue", lty = 2)
225 legend("topright", legend = c("Forecast", "Historical Mean"), col =
      c("red", "blue"), lty = c(1, 2))
226
227 diff_year1 <- year1 - historical_seasonal[,2]
228 diff_year2 <- year2 - historical_seasonal[,2]
229 t_test_year1 <- t.test(diff_year1)
230 t_test_year2 <- t.test(diff_year2)
231 res_acf <- acf(forecast_24m$pred, main = "ACF of Forecasts")
232 pred_range <- apply(rbind(forecast_24m$pred - 1.96*forecast_24m$se,
      forecast_24m$pred + 1.96*forecast_24m$se), 2, diff)
233 cat("Difference Year 1:", round((mean(year1) - mean(ts_rain))/mean(
      ts_rain)*100, 1), "%\n")

```

```

234 cat("Difference Year 2:", round((mean(year2) - mean(ts_rain))/mean(
    ts_rain)*100, 1), "%")
235
236 test_size <- 24
237 train <- window(ts_rain, end = c(end(ts_rain)[1]-2, 12))
238 test <- window(ts_rain, start = c(end(ts_rain)[1]-1, 1))
239 fit_train <- arima(train, order = c(2,1,1), seasonal = list(order =
    c(1,1,1), period = 12))
240 test_forecast <- predict(fit_train, n.ahead = test_size)
241 errors <- test - test_forecast$pred
242 mae <- mean(abs(errors))
243 rmse <- sqrt(mean(errors^2))
244 plot(test, main = "Actual vs Predicted Values")
245 lines(ts(test_forecast$pred, start = start(test), frequency = 12),
    col = "red")

```

Listing 3: Exercise 3 - Time Series Analysis with AR, ARIMA and Seasonal Decomposition

```

1 # 3.1 Mortality forecasting with AR(2)
2
3 library(astsa)
4
5 # Create the time series
6 ts_cmort <- ts(cmort, frequency = 52)
7
8 # Fit AR(2)
9 ar2_fit <- arima(ts_cmort, order = c(2,0,0))
10
11 # 4-week forecast
12 forecast_horizon <- 4
13 forecast <- predict(ar2_fit, n.ahead = forecast_horizon)
14
15 # 95% confidence intervals
16 upper_bound <- forecast$pred + 1.96 * forecast$se
17 lower_bound <- forecast$pred - 1.96 * forecast$se
18
19 # Plot results
20 n <- length(ts_cmort)
21 weeks <- (n - 30):(n + forecast_horizon)
22 plot(weeks[1:31], ts_cmort[(n-30):n], type = "l", lwd = 2,
23      ylim = range(ts_cmort[(n-30):n], upper_bound, lower_bound),
24      xlim = range(weeks),
25      main = "AR(2) Mortality Forecast",
26      xlab = "Week", ylab = "Mortality")
27 forecast_weeks <- (n + 1):(n + forecast_horizon)
28 lines(forecast_weeks, forecast$pred, col = "red", lwd = 2, type = "
    b", pch = 16)
29 lines(forecast_weeks, upper_bound, col = "blue", lty = 2, lwd = 2)
30 lines(forecast_weeks, lower_bound, col = "blue", lty = 2, lwd = 2)
31 legend("topright",
32      legend = c("Historical", "AR(2) Forecast", "95% Confidence

```

```

Interval"),
33     col = c("black", "red", "blue"),
34     lty = c(1, 1, 2), pch = c(NA, 16, NA), lwd = c(2,2,2), bty =
        "n")
35
36 # Residual diagnostics
37 resid_ar2 <- residuals(ar2_fit)
38 par(mfrow = c(2,3))
39 plot(resid_ar2, type = "l", main = "AR(2) Residuals", ylab = "
    Residuals")
40 abline(h = 0, col = "red", lty = 2)
41 acf(resid_ar2, main = "ACF Residuals")
42 pacf(resid_ar2, main = "PACF Residuals")
43 hist(resid_ar2, breaks = 20, col = "lightblue", main = "Residual
    Histogram")
44 qqnorm(resid_ar2); qqline(resid_ar2, col = "red")
45 plot(weeks[1:31], ts_cmort[(n-30):n], type = "l", lwd = 2, main = "
    Latest Values", xlab = "Week", ylab = "Mortality")
46 par(mfrow = c(1,1))
47
48 # Ljung-Box test
49 lb <- Box.test(resid_ar2, lag = 10, type = "Ljung-Box")
50 cat("Ljung-Box test p-value:", lb$p.value, "\n")
51 cat("Coefficients:\n"); print(ar2_fit$coef)
52 cat("AIC:", ar2_fit$aic, "\n")
53
54
55 # 3.2 Global temperature analysis and forecasting with ARIMA
56
57 library(astsa)
58 data(gtemp_land)
59 years <- 1850:2023
60 x_ts <- ts(gtemp_land, start = 1850, frequency = 1)
61
62 # Original plot
63 plot(x_ts, main = "Global Land Temperature (1850-2023)", ylab = "
    Anomaly (degC)", xlab = "Year")
64
65 # ACF/PACF of the original series and first difference
66 par(mfrow = c(1,2))
67 acf(x_ts, main = "ACF of Original Series")
68 pacf(x_ts, main = "PACF of Original Series")
69 par(mfrow = c(1,1))
70 dx_ts <- diff(x_ts)
71 plot(dx_ts, main = "First Difference of Series", ylab = "Difference
    from Previous Year", xlab = "Year")
72 par(mfrow = c(1,2))
73 acf(dx_ts, main = "ACF of First Difference")
74 pacf(dx_ts, main = "PACF of First Difference")
75 par(mfrow = c(1,1))
76

```

```

77 # ARIMA(1,1,1) model
78 fit <- arima(x_ts, order = c(1,1,1))
79 resid_fit <- residuals(fit)
80
81 # Residual diagnostics
82 par(mfrow = c(2,2))
83 plot(resid_fit, main = "ARIMA Residuals", ylab = "Residuals")
84 acf(resid_fit, main = "ACF of Residuals")
85 pacf(resid_fit, main = "PACF of Residuals")
86 hist(resid_fit, main = "Residual Histogram", col = "lightblue")
87 par(mfrow = c(1,1))
88
89 # Ljung-Box test
90 lb_test <- Box.test(resid_fit, lag = 20, type = "Ljung-Box")
91 cat("Ljung-Box p-value:", lb_test$p.value, "\n")
92 cat("AIC:", fit$aic, "\n")
93 cat("Coefficients:\n")
94 print(fit$coef)
95
96 # Forecast for 10 years
97 n_forecast <- 10
98 pred <- predict(fit, n.ahead = n_forecast)
99 z <- qnorm(0.975)
100 upper <- pred$pred + z * pred$se
101 lower <- pred$pred - z * pred$se
102 last_year <- end(x_ts)[1]
103 forecast_years <- (last_year + 1):(last_year + n_forecast)
104 forecast_years <- c(last_year, forecast_years)
105 forecast_pred <- c(tail(x_ts, 1), pred$pred)
106 upper <- c(tail(x_ts, 1), upper)
107 lower <- c(tail(x_ts, 1), lower)
108
109 # Visualization
110 plot(window(x_ts, start = 2000), type = "o", pch = 16, col = "blue"
111      ,
112      xlim = c(2000, last_year + n_forecast),
113      ylim = range(c(x_ts, lower, upper), na.rm = TRUE),
114      main = "Global Land Temperature Forecast",
115      xlab = "Year", ylab = "Anomaly (degC)")
116 lines(forecast_years, forecast_pred, type = "o", col = "red", pch =
117       18, lty = 2)
118 lines(forecast_years, upper, col = "darkgreen", lty = 3)
119 lines(forecast_years, lower, col = "darkgreen", lty = 3)
120 polygon(c(forecast_years, rev(forecast_years)), c(upper, rev(lower)
121           ),
122         col = rgb(0, 0.5, 0, 0.2), border = NA)
123 legend("topleft", legend = c("Historical", "Forecast", "95% CI"),
124       col = c("blue", "red", "darkgreen"),
125       lty = c(1, 2, 1), pch = c(16, 18, NA), bg = "white")
126

```

```

125 # 3.3 Seasonal analysis and trend of water sales
126
127 # Enter data
128 sales_data <- data.frame(
129   Year = c(2016, 2017, 2018, 2019),
130   Winter = c(1, 2, 2, 1),
131   Spring = c(3, 2, 4, 3),
132   Summer = c(6, 7, 8, 8),
133   Autumn = c(4, 5, 5, 6)
134 )
135
136 # Convert to time series
137 sales_ts <- ts(c(t(sales_data[, -1])), start = 2016, frequency = 4)
138
139 # Calculate seasonal indices
140 decompose_sales <- decompose(sales_ts, type = "multiplicative")
141 seasonal_indices <- decompose_sales$seasonal[1:4]
142 cat("Seasonal Indices:\n")
143 print(round(seasonal_indices, 3))
144
145 # Deseasonalization
146 deseasonalized_sales <- sales_ts / seasonal_indices
147
148 # Trend line
149 time_points <- time(sales_ts)
150 trend_model <- lm(deseasonalized_sales ~ time_points)
151 trend_line <- ts(fitted(trend_model), start = 2016, frequency = 4)
152
153 # Cyclical component
154 cyclic_component <- deseasonalized_sales / trend_line
155
156 # Visualization
157 par(mfrow = c(2,2))
158 plot(sales_ts, main = "Raw Sales", ylab = "Sales (million bottles)"
159   , col = "blue")
159 legend("topleft", "Raw", col = "blue", lty = 1)
160 plot(deseasonalized_sales, main = "Deseasonalized Sales", ylab = "
161   Sales", col = "red")
161 lines(trend_line, col = "darkgreen", lwd = 2)
162 legend("topleft", c("Deseasonalized", "Trend"), col = c("red", "
163   darkgreen"), lty = c(1,1))
163 plot(cyclic_component, main = "Cyclical Variation", ylab = "
164   Relative Residuals", col = "purple")
164 abline(h = 1, col = "gray", lty = 2)
165 legend("topleft", "Cyclical", col = "purple", lty = 1)
166 barplot(seasonal_indices, names.arg = c("Winter", "Spring", "Summer
167   ", "Autumn"),
168   main = "Seasonal Indices", ylab = "Index")
169 par(mfrow = c(1,1))

```